SF-16 Synchronous FIFO

Section 3 · Pins and functional operation

3.1 Configuration

Data width 8 bits · 16 entries · almost-full level 14 · occupancy counter 5 bits

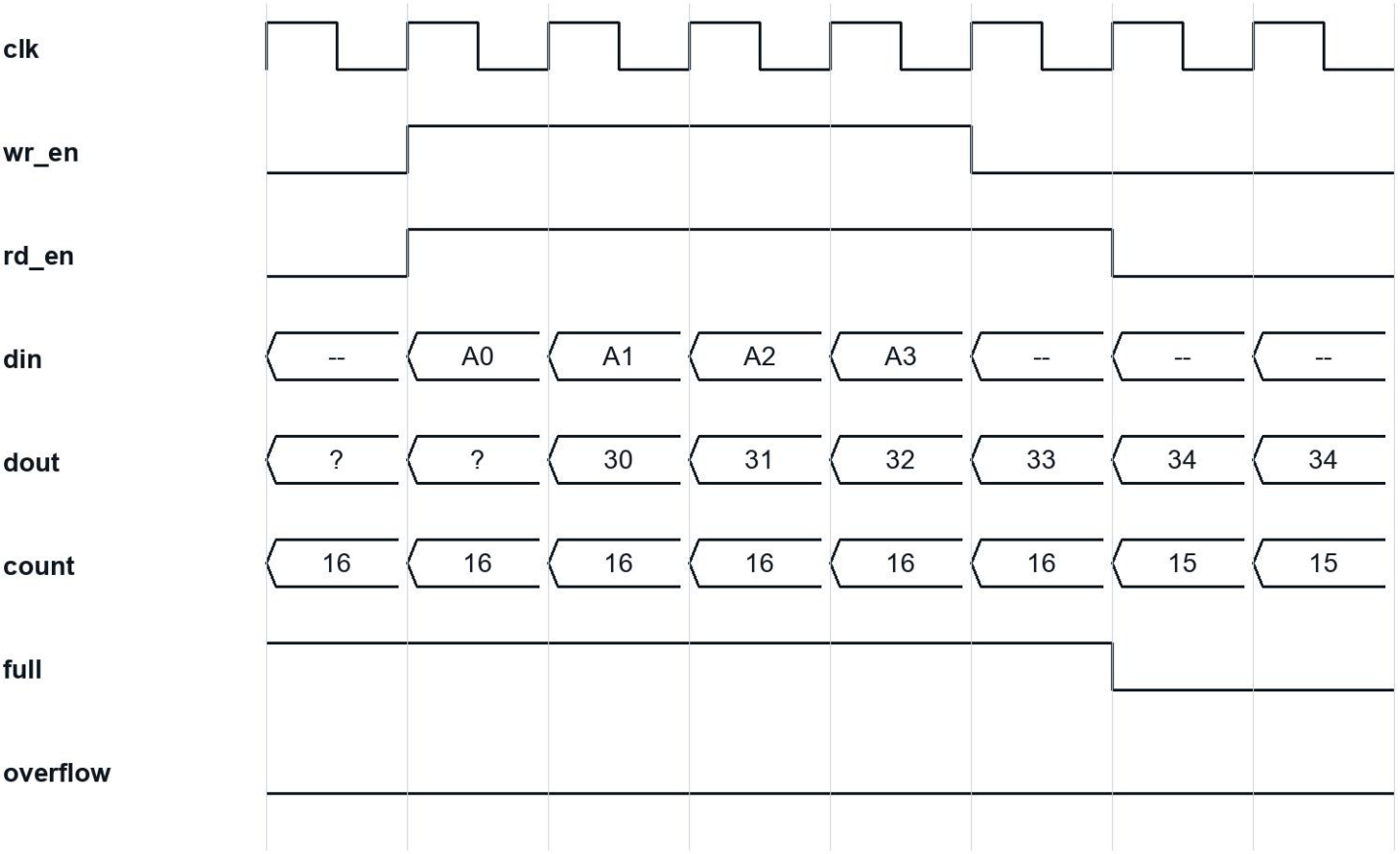
3.2 Pins

Pin	Dir	Description
clk	in	Clock. All state changes on the rising edge.
rst	in	Synchronous reset, active high.
wr_en	in	Push request.
din[7:0]	in	Push data.
rd_en	in	Pop request.
dout[7:0]	out	Pop data, registered (valid after the edge).
full / empty	out	Occupancy is 16 / occupancy is 0.
almost_full	out	Occupancy at or above the almost-full level.
overflow	out	Push refused, high for one clock.
underflow	out	Pop refused, high for one clock.
count[4:0]	out	Words currently held.

3.3 Operation at each rising clock edge

Condition	Result
rst = 1	Storage emptied. count 0, empty 1, dout 00h, overflow 0, underflow 0.
push, not full	din stored; count + 1.
push while full, no pop	Push dropped. overflow pulses for one clock.
pop, not empty	Oldest word appears on dout; count - 1.
pop while empty	Ignored; dout keeps its old value. underflow pulses for one clock.
push + pop while full	Oldest word out, new word in; count stays 16. No overflow.
push + pop while empty	Word stored, pop ignored, underflow pulses.

Figure 4.1 Push and pop in the same cycle while full



Cycle 1-4: the FIFO holds 16 words; push and pop are both high.
Each edge returns the oldest word on dout and stores din; count stays 16, full stays 1,
overflow stays 0 because a pop frees the slot in the same cycle.
Cycle 5: pop only, count falls to 15 and full drops.

Ordering: words leave in the order they were pushed, including when the internal
address wraps from entry 15 back to entry 0.
almost_full is high whenever count is 14 or more.